JOMO KENYATTA UNIVERSITY

OF

AGRICULTURE AND TECHNOLOGY

BACHELOR'S OF SCIENCE IN OPERATIONS RESEARCH

DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCES

TITLE: FORECASTING SECOND HAND TEXTILE DEMAND AND ASSOCIATED WASTE GENERATION USING RECURRENT NEURAL NETWORKS IN NAIROBI COUNTY.

BOR GROUP 2

Declaration

We declare that this is our original work, and the information provided in this report is a true representation of our effort.

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Declaration by the Supervisor

This research project has been presented for examination with my approval as the appointed supervisor.

Dr. Benjamin Muema

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ABSTRACT

Despite being an important industry sector in our country's economy, the consumerism of second-hand clothing has majorly contributed to environmental pollution in Nairobi County. With rapid growth of the second-hand textile industry and increasing importation of second-hand textile, waste generated has emerged as a significant environmental challenge. This project aims to employ recurrent neural networks to predict demand of second-hand clothes and waste generated. The project aims to create a sustainable second-hand textile industry by inventory management to avoid overstocking or stock out situations. The findings of this research aim to provide insights for policy makers, industry stakeholders and environmental organizations ultimately contributing to a more sustainable textile industry in Nairobi.

CHAPTER ONE: INTRODUCTION

1.1 Background of the Study

In recent years, the second-hand textile industry has grown significantly, especially in Kenya. The demand for second hand clothes continues to rise, as evidenced by the increasing volumes in imports considering Kenya is the largest importer of second-hand clothing in Sub-Saharan Africa. This sector provides affordable clothing options with 91.5% of households purchasing second hand clothes worth less than 1000 shillings and 8.5% purchasing second hand clothes worth more than 1000 shillings. The second-hand textile sector not only provides affordable clothing options for low-income households but also serves as a vital source of employment for many individuals engaged in the trade. This trade has a significant economic impact, with the sector contributing at least 1 billion shillings in revenue per month. In urban areas like Nairobi County, managing this textile waste has become a critical concern due to more volumes of second-hand textiles being imported every other day.

There are growing concerns about the environmental impact of the second-hand clothing trade, particularly the influx of waste synthetic clothing. In Nairobi, the demand for second-hand textiles has led to an exponential increase in textile waste, compounding existing environmental challenges. Inefficient waste management practices and limited recycling infrastructure have exacerbated the issue, leading to the accumulation of unusable garments in landfills and illegal dumpsites. The disposal of such textiles poses significant environmental risks, including soil and water pollution, due to the long decomposition time of synthetic fibers and the potential leaching of hazardous chemicals.

Given the pressing nature of these issues, accurately predicting second-hand textile demand and associated waste generation is essential for the establishment of effective waste management strategies, to create sustainability by reducing waste through inventory management hence avoiding overstocking or stock out situations.

This research proposes the use of Recurrent Neural Networks (RNNs) as a powerful predictive tool for modeling and forecasting second-hand textile demand and waste generation patterns in Nairobi County. RNNs are particularly adept at handling sequential data and can capture intricate temporal dependencies, making them suitable for analyzing trends in consumer behavior over time. By leveraging RNNs to forecast demand, this study aims to provide stakeholders, including policymakers and waste management authorities, with actionable insights that can drive sustainable practices within the second-hand textile industry.

1.2 Statement of the Problem

Kenya imports a substantial amount of second-hand clothes over the years. A significant portion of Nairobi's population lives below the poverty line, making second hand clothing more accessible and affordable hence increasing the demand of second-hand clothing. Despite the high demand for second hand clothing, import patterns often lead to overstocking and unsold inventory (dead stock), much of which eventually becomes waste. With Kenya becoming the largest importer of second-hand clothing in Sub-Saharan Africa, that comes with a significant amount of textile waste leading to landfills, soil pollution, since polyesters take decades to decompose and water pollution, especially the Nairobi river. The lack of proper forecasting tools makes it difficult for businesses and municipalities to anticipate textile demand and prepare for the waste that follows. This unpredictability leads to insufficient resource allocation, increased dumping and pressure on landfills and on the environment. We are using LSTM which is a specialized type of Recurrent Neural Network (RNN) that excels at processing and learning sequential data. It has a flexible memory control through the gating mechanisms which allow information to keep,update or discard.

1.3 Objectives

Main Objective

To use an RNN architecture that forecasts second hand textile demand and associated waste generation in Nairobi County

Specific objectives

i. To identify spikes and trends of the demand and associated waste generated of second hand textile clothing.
ii. To forecast demand of second hand textile clothing and associated waste generated using RNN-Long Short Term Memory neural network.
iii. To evaluate performance of the model using statistical metrics.

1.4 Justification

The second-hand clothing trade has become an increasingly important industry within the informal sector as it provides affordable clothes to Kenyans of all socio-economic classes. The rise of fast fashion in developing countries leads to an oversupply of clothing most of which ends up in developing countries like Kenya. The clothes are imported as bales as some which don't fit, some are dirty and others are in bad conditions to sell and end up being waste. An overwhelming amount of clothing shipped to Kenya is waste synthetic clothing. The environmental impact of textile waste is a growing concern with disposal of discarded clothing in landfills and water bodies raising health risks.

Predicting the demand of second-hand clothing and the associated waste is to avoid overstocking which will also prevent textile waste. This will help implement regulation and policies to control the import of second-hand clothing and create more sustainable and local jobs. Predicting demand and associated waste will help in developing effective strategies for waste management and environmental protection.

A study in 2020 by Enock Gideon Musau investigated the effect of procurement performance practices on waste management in textile manufacturing firms in Nairobi City County. Musau's study is predominantly focused on identifying the relationship and correlations between procurement practices and waste management while our model is designed for forecasting future demand and waste which provides a more proactive approach to textile waste allowing for planning and interventions before surplus and waste becoming a significant issue. A study in China used the LSTM and XGBoost hybrid model to predict textile waste based on economic indicators and purchase behavior (Chen et al., 2022). In contrast, our study aims to reduce overstoking and support urban waste management. The key gap lies in the focus on post consumer second hand textiles and operational decision making in developing urban context rather the consumer driven waste prediction in a developed economy.

This project proposes the use of RNN (Recurrent Neural Networks) to model and forecast textile demand and waste generation patterns in Nairobi County. By leveraging RNNs this study aims to provide stakeholders, including policy makers and waste management authorities with actionable insights that will drive sustainable practices within the second hand textile industry. This will contribute to efficient resource allocation and waste management practices highlighting the potential for economic growth and environmental preservation.

CHAPTER TWO: LITERATURE REVIEW

Introduction

The global surge in textile waste, particularly in urban centers like Nairobi County, has emerged as a critical environmental and public health challenge. With millions of tons of second-hand clothing imported annually into Kenya—reaching 53,979 tons in the second quarter of 2023 alone (KNBS, 2022b)—Nairobi's landfills and waterways are increasingly overwhelmed by synthetic and low-quality textiles.

Theoretical Framework

The growing textile waste crisis underscores the urgent need for predictive modeling to inform sustainable waste management strategies. Nairobi's improved waste collection systems—such as the Clean, Healthy, Wealthy Nairobi initiative (JICA, 2016)—have not translated into circular textile waste solutions, as landfilling remains the default disposal method (Njoroge et al., 2014). These challenges align with critiques of the linear economy model (Ellen MacArthur Foundation, 2015), where inadequate infrastructure and reactive policies perpetuate waste accumulation.

Recent advances in machine learning offer tools to bridge this gap.Models, such as those proposed by Bianchi et al. (2021), demonstrate how integrating socioeconomic factors (e.g., income levels, consumer behavior) with waste data can improve estimation accuracy. Similarly, Wang & Zhang (2020) highlight the efficacy of LSTM neural networks in forecasting fashion waste generation, leveraging their ability to capture nonlinear temporal trends (e.g., seasonal demand spikes, fast-fashion turnover). These approaches address a critical limitation identified by Kumar et al. (2021): traditional statistical models often fail to account for the complex, dynamic interplay of economic and behavioral drivers behind textile waste.

A hybrid LSTM-XGBoost framework (Chen et al., 2022, theoretical) could synthesize these insights. Here, LSTM processes time-series data (e.g., monthly textile imports, GDP fluctuations). This approach aligns with Complex Adaptive Systems (CAS) Theory (Holland, 1995), treating waste streams as dynamic systems requiring adaptive, data-driven interventions. For instance, in Nairobi, predictive tools could optimize collection routes for textile waste or simulate the impact of extended producer responsibility (EPR) laws—addressing the current reliance on landfilling.

Waste Disposal and Environmental Impact

Current waste management practices in Nairobi remain fragmented, with textile waste often lumped into general waste streams due to inadequate categorization and poor public awareness (Mugambi et al., 2020). Changing Market in 2023 found that a large proportion of used clothing is dumped in both official and informal landfills in Nairobi, with significant amounts polluting the Nairobi River and its downstream areas. An assessment of the material composition revealed that much of this waste is synthetic, leading to micro plastic leaching into water and soil and posing severe environmental and health risks to local communities. In some cases, the lowest quality clothing is burned as fuel to roast food, causing residents and waste pickers to inhale toxic smoke that contributes to respiratory illnesses like asthma. With some landfills managed by criminal networks and governmental responsibilities fragmented between national and local authorities amid a growing influx of imported waste, these improperly disposed materials release toxic pollutants—including microplastics and greenhouse gases—thereby exacerbating environmental degradation and health hazards (Changing Markets, 2023; Kibera Muthoni, 2021).

Empirical Review

Despite challenges such as unreliable waste generation data and fluctuating demand for recycled products, initiatives like Takataka Solutions, Africa Collect Textiles, and start-ups like Closing the Loop on Textile Waste in Kenya are working to divert textile waste through recycling and upcycling. For example, Africa Collect Textiles estimates that 150–200 tons of non-reusable textiles are discarded daily in Nairobi, yet their efforts are limited by a lack of predictive capacity for efficient resource allocation. To overcome these challenges, the study recommends that brands collaborate with recycling firms in developing countries like Kenya by providing finances, infrastructure, and international market access for recycled products such as some fabrics, such as T-shirts, are redesigned into polishing and cleaning materials. Fibers obtained from used textiles and carpets are reprocessed into soil reinforcement for enhancing soil strength and stability.

Conceptual Framework

This gap in demand forecasting and waste volume prediction presents an opportunity for advanced computational tools. Recurrent Neural Networks (RNNs), a class of machine learning models adept at analyzing sequential data, have shown promise in waste management contexts. Unlike traditional methods, RNNs particularly Long Short-Term Memory (LSTM) networks can model complex, time-dependent patterns in waste generation, such as seasonal fluctuations in second-hand clothing imports or spikes in post-holiday disposals.

Applied to Nairobi, RNNs could analyze historical import data, consumer behavior trends, and climatic factors (e.g., rainy seasons increasing dumping rates) to forecast textile waste volumes. Such predictions would enable municipalities and recyclers to optimize collection routes, scale recycling operations, and align efforts with circular economy goals.

Furthermore, RNNs could enhance demand prediction for recycled textile products, addressing a key barrier to market stability. By training models on sales data from upcycling startups, social media trends, and regional economic indicators, stakeholders could anticipate demand for items like recycled carpets or repurposed fabrics, reducing overproduction and stockpiling. This approach aligns with the "mali mali" barter system proposed by Africa Collect Textiles, where communities exchange used clothing for recycled goods, fostering a participatory circular economy.

However, the success of such models depends on addressing data scarcity, a challenge highlighted in the Kenyan context. Collaborative data collection, leveraging mobile platforms and partnerships with grassroots organizations, could crowdsource real-time inputs on waste flows and consumer preferences, refining RNN accuracy.

CHAPTER THREE: METHODOLOGY

3.0 Introduction

A recurrent neural network (RNN) is an improved multilayer perceptron network, which includes the input layer, hidden layer, output layer. RNNs perform the same task for every element of a sequence, with the output being dependent on the previous computations. Feedforward neural networks can only map from input to output Fig. 1. An unfolded traditional RNN network vector (one-to-one mapping) whereas RNN can map from the entire history of previous inputs to each output (sequence-to-sequence mapping). A RNN can be thought of as multiple copies of the same network, each passing a message to a successor.Figure 1 gives a simple RNN with one input unit, one output unit, and one recurrent hidden unit unfolded into a full network. X_t is the input at time step, O_t is the output at time step t. S_t is the hidden state at time step, and it is the 'memory' of the network. W, U, V are hidden state weighted matrix, input weighted matrix and output weighted matrix respectively

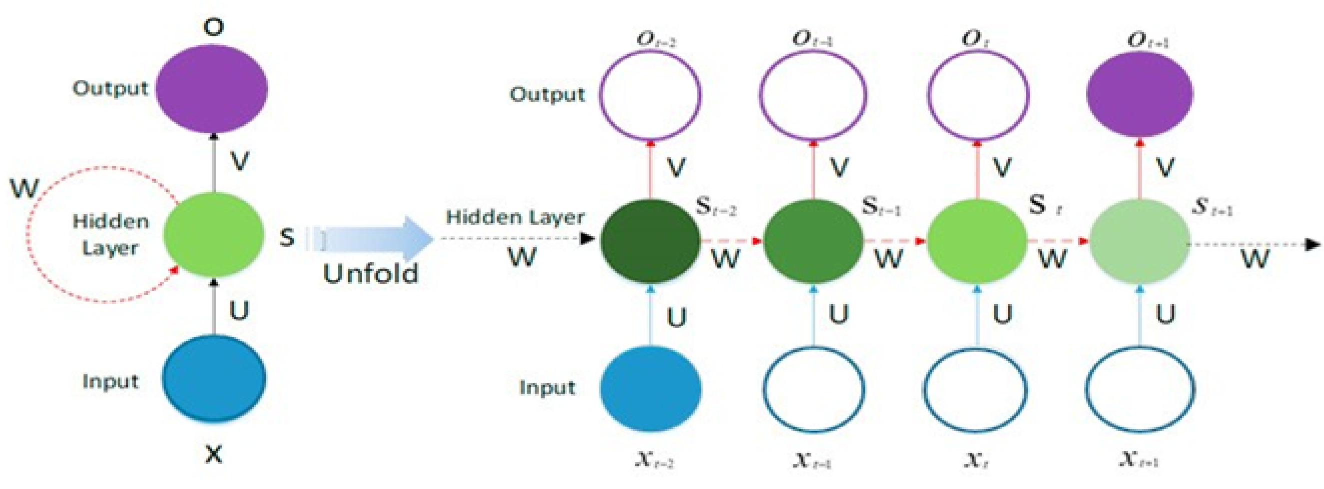


Fig. 1. An unfolded traditional RNN network

The distinctive characteristic lies in that the output of the hidden layer with the present information is transferred to the hidden layer of the next time step as part of the input. This kind of loop can preserve the information of the previous step to keep the data dependency, which improves the ability of learning and abstracting from the sequential data. However, the vanishing gradient problem which makes it hard for the network to learn from previous inputs because updates are too tiny during the calculation of the back-propagation makes the influence of the input cannot be far transferred; in other words, the performance of RNN is not satisfying for long-term dependence problems.

To address the problem of vanishing gradients, we use Long Short Term Memory to control the flow of information and keep the gradient stable during the predicted frame.

3.1. Long Short Term Memory

LSTM networks address the problem of vanishing gradients of RNN by splitting in three inner-cell gates and build so-called memory cells to store information in a long range context . A typical LSTM NN cell is configured mainly by four gates: input gate, input node, forget gate and output gate. Input gate takes a new input point from outside and process newly coming data. Memory cell input gate takes input from the output of the LSTM NN cell in the last iteration. Forget gate decides when to forget the output results and thus selects the optimal time lag for the input sequence. Output gate takes all results calculated and generate output for the LSTM NN cell

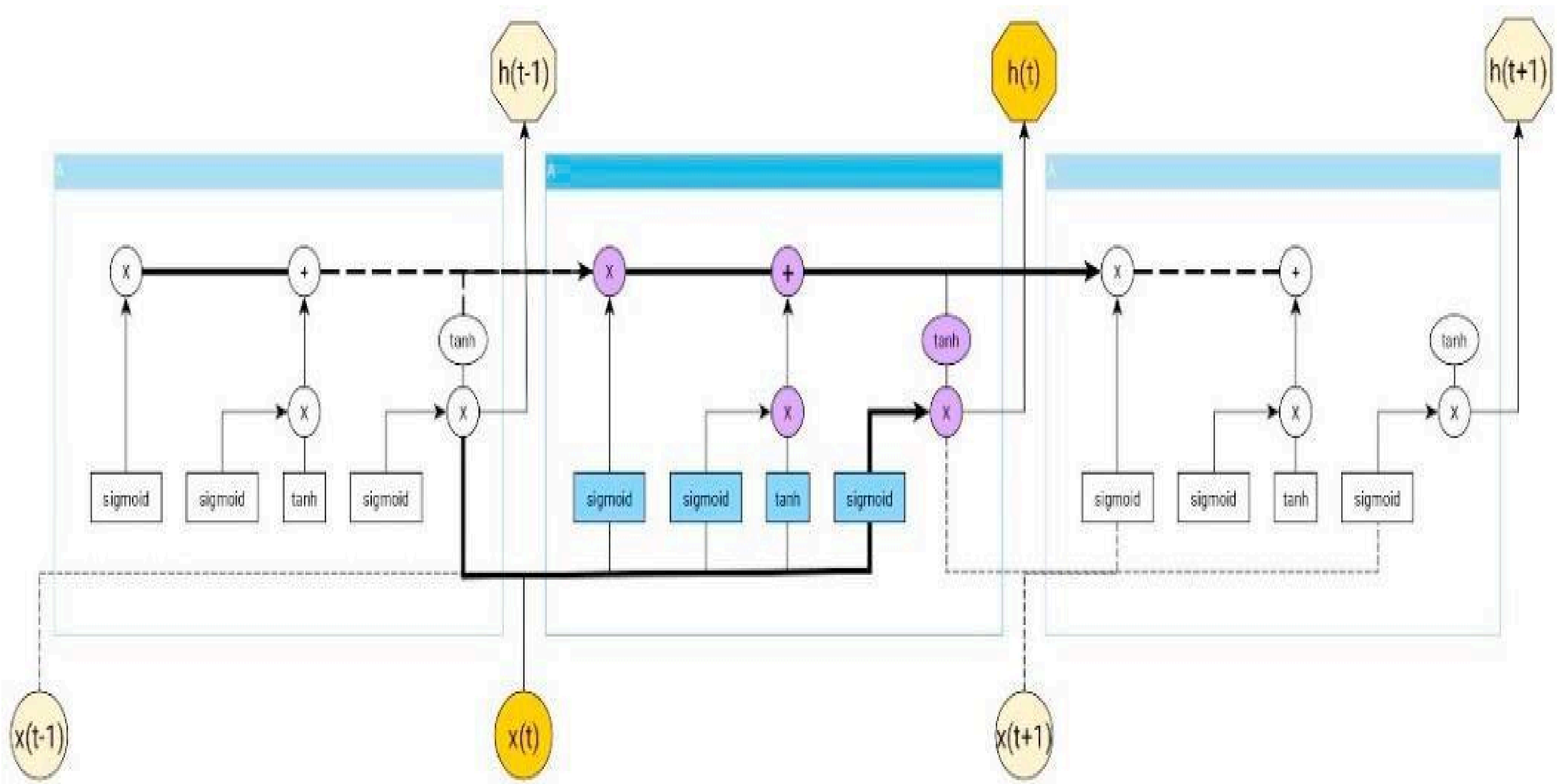


Fig. 2 The typical structure of LSTM NN cells

In our textile demand and associated waste generation prediction model, a linear regression layer is applied on the output layer of the LSTM cell.

Let us denote the input time series as $X = (x, x_2, \dots, x_T)$------(1)

hidden state cells as $H = (h_1, h_2, \dots h_T)$------(2)

output sequence as $Y = (y_1, y_2, \dots, y_T)$------(3)

LSTM NNs do the computation as follows:

Hidden state is calculated as: $h_t = \sigma(W_{hy}x_t + W_{hh}h_{tt-1} + b_h)$------(4)

The output is calculated as: $y_t = W_{hy}h_t + b_y$ —----(5)

Where t=1,…,T

The LSTM structure depicted above is implemented through the

following functions:

$it = \sigma(Wxixt + Whi\, ht-1 + Wcict-1 + bi)$ —-------(6)

$ft = \sigma(Wxfxt + Whf\, ht-1 + Wcfct-1 + bf)$--------(7)

$c_t = f_t * c_{t-1} + i_t * \tan h\, (W_{xc}x_t + W_{hc}h_{t-1} + b_c)$---------(8)

$ot = \sigma(Wxoxt + W_{ho}h_{t-1} + Wcoct + bo)$------------(9)

$h_t = o_t \tan h\, (c_t)$ —-----(10)

Where t=1,…,T

i, f, o and c denote the mentioned inner-cell gates, respectively the input gate, forget gate, output gate, and cell activation vectors.

To measure how well the LSTM network performs, we employed a loss function tailored to our forecasting task. For this research project, we used Mean Absolute Error (MAE) as the primary loss function due to its interpretability and robustness to outliers and stable to noisy data giving more realistic average error. MAE measures the average absolute difference between predicted and actual values, providing a direct and intuitive estimate of model error. The Mean Absolute Error is calculated as:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

- y_i: actual value
- $\hat{y}_i$: predicted value
- n: number of samples

3.2. Data Collection

The data used in this study was compiled from authoritative sources and public datasets spanning 2000 to 2023. Key data points include inflation rates, import volumes, cost per ton, textile waste generated, average annual temperature, annual rainfall, and significant events affecting textile waste management in Kenya.

Secondary sources for economic and environmental data include the Kenya National Bureau of Statistics (KNBS) specifically the Economic Survey and Statistical Abstracts which provided import data,textile waste generated and cost information. Inflation data was cross-referenced with World Bank Open Data

Environmental metrics, such as average temperature and rainfall, were obtained from Africa Data Hub to ensure accuracy and completeness.. Qualitative data on key events such as policy changes, economic crises, and environmental interventions were manually curated from news archives, government reports, and NGO publications to provide context for observed trends.

3.3. Data Analysis

3.3.1.Tools and Libraries

The data analysis was conducted using **Python** programming

language, leveraging powerful libraries such as:

- **Pandas** for data manipulation
- **NumPy** for numerical operations and efficient handling of array-based data
- **Matplotlib & Seaborn** for visualization
- **Scikit-learn** for preprocessing
- **TensorFlow & Keras** for building and training the LSTM model

3.3.2. Preprocessing Steps

To prepare the data for modeling:

- **Normalization:** All numerical features were scaled using **MinMaxScaler** to ensure uniformity and accelerate model convergence.

- **Categorical Encoding:** Key event-based textual annotations (e.g., policy changes or trade bans) were encoded using **LabelEncoder**.
- **Sliding Window Generation:** Time-series sequences were structured using 3-month rolling windows to allow the model to learn temporal dependencies effectively.

3.3.3. Model Architecture and Training

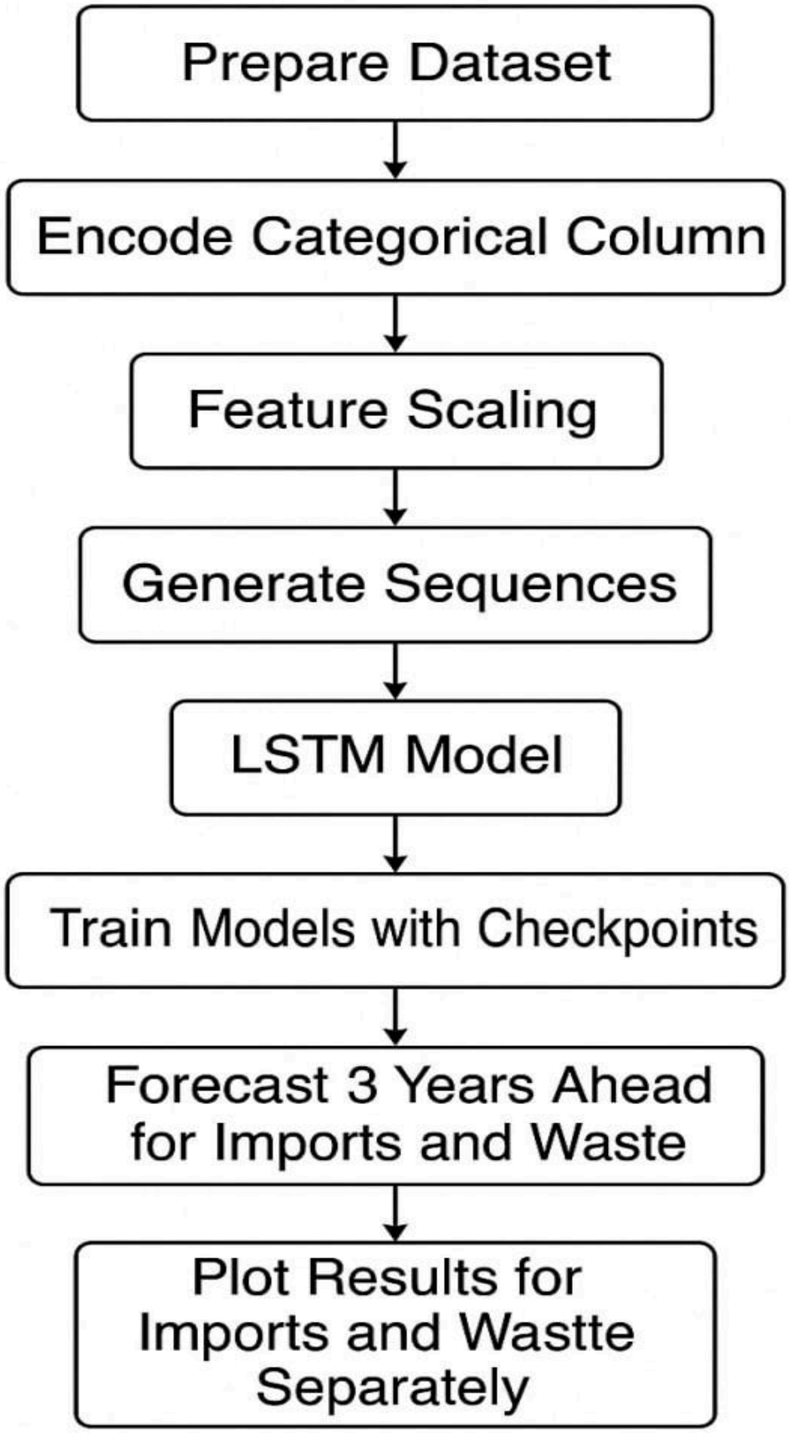
A **stacked LSTM model** was implemented with the following architecture:

- **Four hidden LSTM layers**
- **Dropout regularization** applied between layers to reduce overfitting
- **Mean Squared Error (MSE)** used as the loss function
- The model was trained to predict **yearly** imports and waste levels based on preceding 3-month sequences

3.3.4.Model Evaluation Summary

The LSTM model's performance was assessed using three key metrics:

- **Mean Squared Error (MSE):** Measures the average squared difference between predicted and actual values. Lower MSE values indicate higher accuracy.
- **Mean Absolute Percentage Error (MAPE):** Shows the average prediction error as a percentage. It helps understand model performance in relative terms, with lower percentages indicating better accuracy.
- **R-squared (R^2):** Indicates how well the model explains the variability in the data.


Prepare Dataset
Encode Categorical Column
Feature Scaling
Generate Sequences
LSTM Model
Train Models with Checkpoints
Forecast 3 Years Ahead for Imports and Waste
Plot Results for Imports and Wastte Separately

3.4.RESULTS.

The predictive models developed for analyzing textile waste and import trends yielded the following performance metrics:

3.4.1.Forecasted Textile Imports and Waste (2024–2026):

Year	Imports (tons)	Textile Waste (tons)
2024	24116	7887
2025	23351	7087
2026	23521	7954

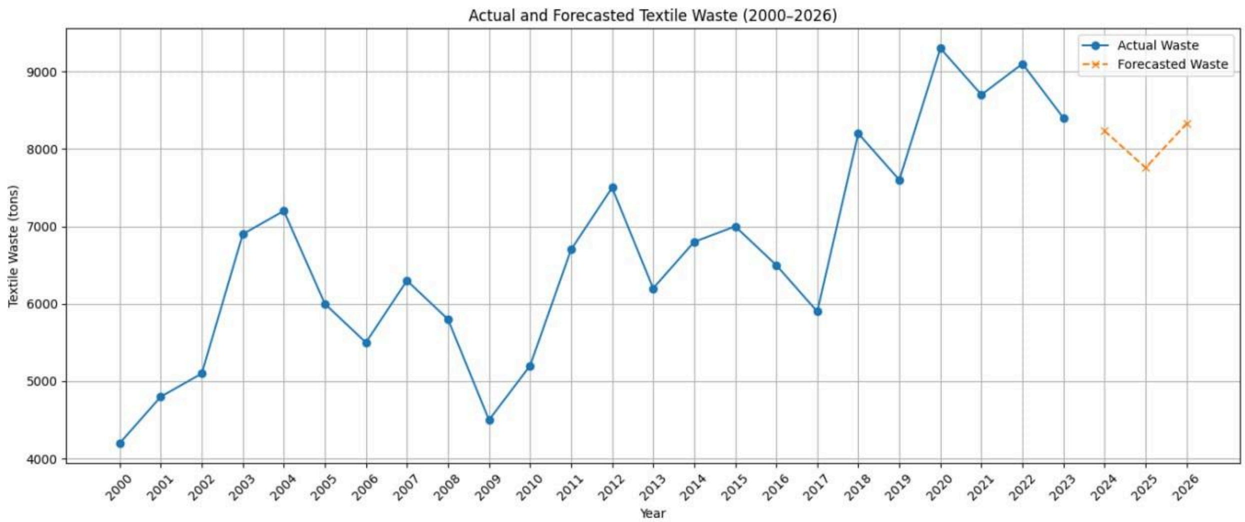
3.4.2. Imports Model:

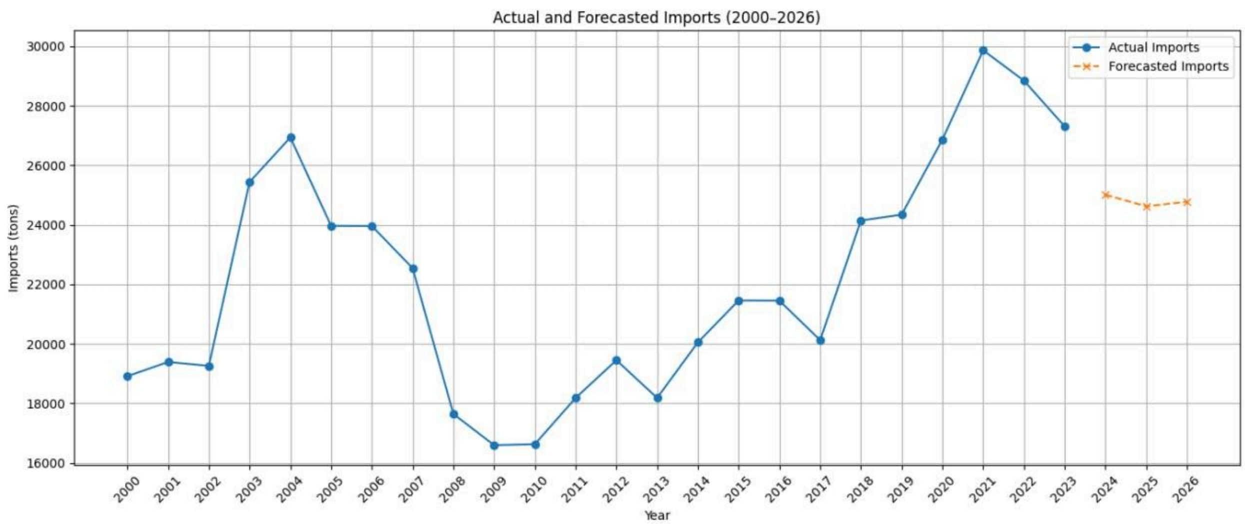
- **MAE (631.07 tons):** On average, predictions were off by 624 tons.
- **MAPE (2.94%):** The model's error is 2.94% of actual values, indicating high accuracy.
- **R^2 (0.9512):** The model explains 95.1% of the variation in import data a very strong fit.

3.4.3.Textile Waste Model:

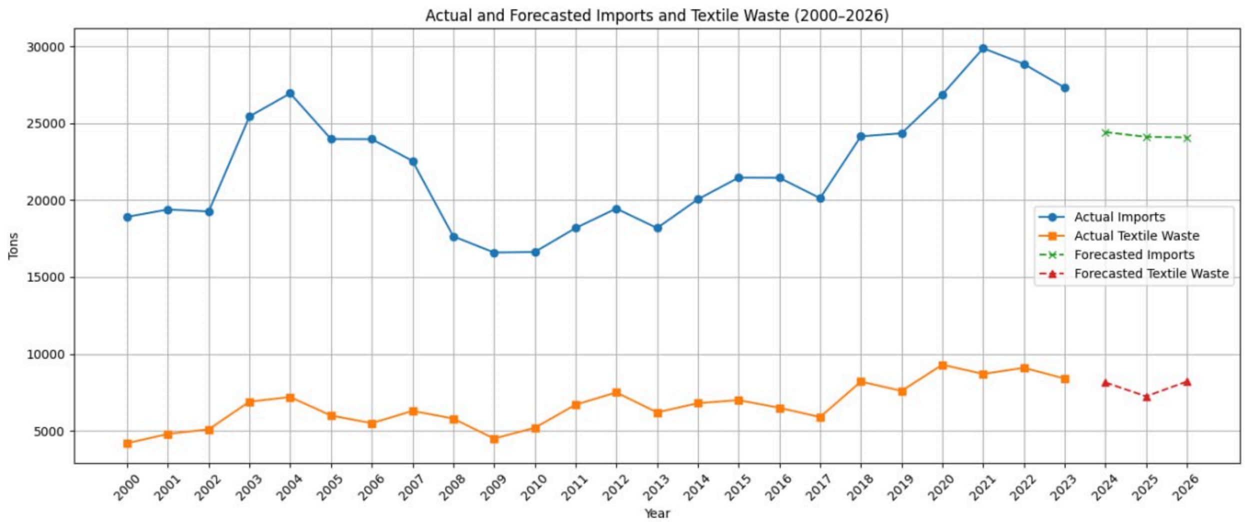
- **MAE (218.07tons):** Predictions missed actual values by about 428 tons on average.
- **MAPE (3.02%):** The model has slightly higher error, but still within a good range.
- **R^2 (0.9415):** Explains 94.2% of the variation in textile waste fairly accurately.

We fit an RNN model(Long Short Term Memory) that identifies spikes and trends of the demand and waste generated.





We forecasted demand for second hand textile and associated waste using LSTM model for the next 3 years.



CHAPTER FIVE: CONCLUSION AND RECOMMENDATIONS

5.1.CONCLUSION

In this study, we used LSTM which is a type of an RNN to forecast the demand of second hand textile and associated waste.By providing accurate demand forecasts, the model can help reduce overstocking, improve how second-hand textiles are sorted and processed, and lower the environmental impact of textile waste.The insights derived from this model can help stakeholders in the second-hand textile sector—including importers, sorters, and waste managers—to make more informed decisions, reduce overstocking, and plan for more sustainable waste management strategies.

Future research can improve this model by including other factors like economic conditions, population changes, or weather patterns. Adding this model to decision-making tools can also support more data-driven and sustainable policies in the textile industry.

5.2.RECOMMENDATIONS

1.**Develop Policy Based on Forecasting Trends.**
With projections indicating a steady increase in waste, especially from textiles, proactive policies are essential. This includes regulating second-hand clothing imports and promoting domestic recycling initiatives to reduce landfill pressure and unethical disspossing.

2**. Promote Data-Driven Decision-Making for Sustainable Practices.**
policy makers should encourage the adoption of data-driven decision making for managing textile waste, backed by models like RNN's to help municipalities and businesses understand and control waste patterns. By using predictive models to estimate future textile waste, stakeholders can better prepare for recycling, repurposing, or redistribution programs, reducing the environmental impact of landfill dumping.

3.**Policy recommendations to import tariffs based on demand forecasting.**
Adds regulatory control to an informal system.

4. **Implement dynamic import schedule (monthly).**
Monthly import instead of bulk imports will reduce textile oversupply which in turn lowers the waste levels.

References

Changing Markets Foundation. (2023). Trashion: The stealth export of waste plastic clothes to Kenya. Retrieved from https://changingmarkets.org/portfolio/fossil-fashion

Bestseller Foundation. (2023). TakaTaka Textile Recycling. Retrieved from https://bestseller.org/takataka-textile

Africa Collect Textiles. (2021). *Africa Textile Collect - Home, Africa Collect Textiles*. Retrieved from https://africacollecttextiles.com/

KNBS. (2022b). *[Please provide the specific title of the economic survey or report where this data is found]*. Kenya National Bureau of Statistics. [If available, provide the URL of the report]

Africa Collect Textiles. (2021). *Africa Textile Collect - Home, Africa Collect Textiles*. Retrieved from https://africacollecttextiles.com/

Amasuomo, E. (2016, November). The concept of waste and waste management. Journal of Management and Sustainability, 6(4), 88.

https://doi.org/10.5539/jmi.v6n4p88

Aishwariya, S., & Jaisri, M. J. (2020). *Harmful effects of textile wastes*. Fibre2Fashion, p. 8. Retrieved from

Kenya

https://www.fibre2fashion.com/industry-article/8696/harmful-effects-of-textile-waste

Gakungu, N., Gitao, A. N., & Kimani, B. (2012, January). Solid waste management in . Environmental Studies,43, 45

Chen, X., Li, Y., & Wang, Z. (2022). Predicting textile waste using a hybrid LSTM and XGBoost model based on economic indicators and purchase behavior. Journal of Waste Management Research, 15(3), 456–470

***Bianchi, F., et al. (2021).** "A hybrid machine learning approach for textile waste estimation based on socioeconomic factors." Waste Management, 126, 735-745.*

***Wang, L., & Zhang, G. (2020).** "Forecasting fashion waste generation using LSTM neural networks." Resources, Conservation & Recycling, 158, 104798.*

***Kumar, S., et al. (2021).** "Textile waste prediction models: A review of statistical and machine learning approaches." Journal of Cleaner Production, 279, 123686.*

Leaders for a Better World. (n.d.). Smart fashion production: Predicting demand to minimize waste. Retrieved April 13, 2025, from https://leadersforabetterworld.net/smart-fashion-production-prediciting-demand-to-minimize-waste

Wikipedia contributors. (n.d.). Waste input-output model. Wikipedia. Retrieved April 13, 2025, from https://en.wikipedia.org/wiki/Waste_input-output_model